An Understanding Kit

Eight results from the corpus, to rebuild by hand

Patrick S. Mahon

October 2026

How to use this. Read one problem's statement. Close the PDF and work on paper until you see *why* it must be true. That means rebuilding it, not verifying it. Hints are in Part II and proofs in Part III, deliberately far away. The problems run from a few minutes to an hour. None needs anything outside this document. A problem that lights up tells you something; one that doesn't tells you something too.

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Part I
Problems

Shared definitions

Definition 1 (two-valued state)**.** For a family $\mathcal{D}$ of subsets of a set Ω closed under complements and countable disjoint unions, a *two-valued state* is $s : \mathcal{D} \to \{0, 1\}$ with $s(\Omega) = 1$ that is additive on disjoint pairs. It is *σ-additive* if additivity holds for countable disjoint families, and *finitely additive* if only for finite ones. The *Dirac* δ_ω is $s(E) = 1 \iff \omega \in E$.

Definition 2 (language, two tokens)**.** A *language* is a relation $\rho \subseteq A \times A$ on a finite alphabet A, read as a digraph. It is *primitive* if some power of its adjacency matrix is entrywise positive (equivalently: strongly connected, and the gcd of its cycle lengths is 1). *Two tokens* sit on distinct vertices and move *synchronously*: at each step each token crosses one arc of ρ. They may never occupy the same vertex at the same time.

1 Tiny

Problem 1 (stationary two-valued states are fixed points)**.** *Let A be countable and $\Omega = A^{\mathbb{Z}}$ with the product σ-algebra.*

(a) Every σ-additive two-valued state on Ω is a Dirac δ_ω.

(b) If it is also shift-invariant ($s(S^{-1}E) = s(E)$), then ω is a constant sequence.

Why it matters. "Two-valued" means *one trajectory*, not "deterministic dynamics". So a genuine stationary process is never two-valued. This is why the σ-essential lane sits off the path to dynamics.

Source: not written elsewhere; derived 2026-10-09 in conversation.

Problem 2 (out-of-phase tokens never collide)**.** *Let $\lambda : R \to \mathbb{Z}_c$ be a function on a set of vertices. Token 1 is only ever at $w \in R$ at times $t \equiv \lambda(w)$, and token 2 only at times $t \equiv \lambda(w)+e \pmod{c}$, with $e \not\equiv 0$. Show they never collide in R.*

Why it matters. This is the engine of the reach-lemma attack. A phase difference is a collision-proof shield.

Source: `papers/reconstruction/notes/joint2_wielandt_finding.md`, "(2) Out-of-phase tokens never collide".

Problem 3 (the ascending ring)**.** *Fix $L \geq 2$ and let $C_k = \{x \in \mathbb{N}^L : x_k < x_{k+1}\}$, indices mod L.*

(a) No point x lies in every C_k.

(b) Suppose that, for each window $\{k, k+1\}$, s_k is a σ-additive two-valued state on the σ-algebra of events depending only on (x_k, x_{k+1}). Suppose also that adjacent windows agree on events of their shared coordinate, and that $s_k(C_k) = 1$ for all k. Show this is impossible.

Why it matters. Drop "σ-" in (b) and the configuration *exists*: an ordered product (standard name: tensor product of ultrafilters, Hindman–Strauss) ultrafilter makes every window an ascent. That finitely additive object, glued on a carrier, is the Lean-checked countable σ-essential witness (every $L \geq 4$).

Source: `notes/open_questions/delay_embedding/collision_vs_generation.md` (update 2026-10-09); Lean `formalization/QuerySL_sigmaEssential`.

2 Small

Problem 4 (the cycle lemma)**.** *Let ρ be primitive and $C = (x_0, \dots, x_{c-1})$ a simple cycle, $c \geq 2$. Two tokens* chase *on C at offset $j \neq 0$: both follow C, token 2 being j positions ahead. Show that:*

(a) from any offset, by legal non-colliding moves and back again, the tokens can reach a chase at any other offset;

(b) among the closed two-token runs so produced there are lengths with gcd 1.

Why it matters. This is the reason chases are *aperiodic*. Aperiodicity is exactly what the flagship's forcing lemma needs.

Source: `joint2_wielandt_finding.md`, "CYCLE LEMMA (proved 2026-10-09)".

Problem 5 (one chase class)**.** *Let C, C' be simple cycles of length ≥ 2 sharing a vertex v. Find a single synchronous step that turns some chase on C into a chase on C'. Conclude, using that the simple cycles of a strongly connected digraph are chained by shared vertices, that all chases on all cycles are mutually reachable.*

Source: `joint2_wielandt_finding.md`, "LEMMA K*".

Problem 6 (no witness in the Boolean case)**.** *Let $\mathcal{L}$ be a Boolean σ-algebra on Ω and $B \subseteq \mathcal{L}$ finite and closed under complement. Let s_0 be a two-valued assignment on B that extends to* some *finitely additive two-valued state μ on $\mathcal{L}$. Show s_0 extends to a Dirac.*

Why it matters. This is the baseline the whole σ-essential lane pushes against. Coherence supplies μ, and distributivity turns μ into a point. A witness must break the second step.

Source: `papers/sigma_essential/sigma_essential_body.tex`, Prop. `prop:boolean`.

3 Medium

Problem 7 (Dirac test and localization)**.** *With B finite and complement-closed and s_0 two-valued on B, put $K(s_0) = \bigcap\{E \in B : s_0(E) = 1\}$.*

(a) δ_ω extends s_0 iff $\omega \in K(s_0)$.

(b) Hence a pattern that is finitely coherent but has no σ-additive extension needs $K(s_0) = \varnothing$ and *must kill every non-Dirac σ-additive extension. Explain why the second clause is where all the difficulty lives.*

Source: `sigma_essential_body.tex`, Lemma `lem:dirac-iff`, Thm `thm:localization`, Remark `rem:localization-scope`.

Problem 8 (the parity theorem, odd half)**.** *The golden-mean language on $\{0,1\}$ forbids 11. Observe it around a ring of length L through the L adjacent-pair windows. Let each window assign probability $\frac{1}{2}$ to each of 01 and 10.*

(a) The windows agree on shared sites.

(b) For odd L, no probability measure on legal ring configurations has these window marginals.

(c) Stretch. *For even L every coherent datum is realisable. (Hint section only.)*

Why it matters. This is the flagship's smallest real instance of contextuality, and it is determined by parity.

Source: `papers/reconstruction/notes/archive/reconstruction_body_PROVED_ARCHIVE.tex`, § "The parity theorem".

For comparison. Re-read your own CE / Stone route (`papers/paper_i/probability_from_observation_bod` "Stone realization from finitely additive data") and notice what the click felt like.

Part II

Hints

1. (a) For each time k, the sets $\{x_k = a\}$, $a \in A$, are a countable partition. How many can be true? (b) What is $s \circ S^{-1}$ for a Dirac?

2. Compare the two congruences at a hypothetical meeting.

3. (a) Follow the inequalities around. (b) Show each s_k is a Dirac at some pair, then glue.

4. An *excursion* is a walk $x_i \to x_{i'}$ whose interior avoids C. While one token takes it, the other stays on C. Track the offset. Which quantity changes it? Why do these quantities generate $\mathbb{Z}_c$? (Think about closed walks at x_0 and primitivity.) Removing one vertex from a Cayley graph: what keeps it connected?

5. Put token 1 at v and token 2 one step behind v on C.

6. The finite Boolean algebra generated by B has atoms. Exactly one of them is true.

7. (a) Use complements: every false constraint is a true constraint in disguise.

8. (b) If every window is a.s. 01 or 10, what must a global configuration look like? (c) The coherence polytope is the fractional stable-set polytope of the cycle. Look at $\det(I + P)$ for the cyclic shift P.

Part III
Proofs

Problem 1. (a) For each k the events $\{x_k = a\}$ $(a \in A)$ are countably many, disjoint, and cover Ω, so by σ-additivity exactly one, say a_k, has value 1. Let $\omega = (a_k)_k$. Then $\{\omega\} = \bigcap_k \{x_k = a_k\}$, a countable intersection of value-1 events. Its complement is a countable union of value-0 events, which can be disjointified into value-0 events, so it has value 0 and $s(\{\omega\}) = 1$. Hence $s(E) = 1 \iff \omega \in E$. (b) $\delta_\omega \circ S^{-1} = \delta_{S\omega}$. Invariance gives $\delta_{S\omega} = \delta_\omega$, so $S\omega = \omega$, so ω is constant. □

Problem 2. A meeting at w at time t needs $t \equiv \lambda(w)$ and $t \equiv \lambda(w) + e$, so $e \equiv 0$, a contradiction. □

Problem 3. (a) $x_0 < x_1 < \cdots < x_{L-1} < x_0$. Equivalently, take k maximising x_k; then $x_k < x_{k+1}$ contradicts maximality. (b) As in Problem 1(a), each s_k is $\delta_{(a_k, b_k)}$ on its window, since $\mathbb{N}^2$ is countable and the singletons are measurable. Agreement on the shared coordinate gives $b_k = a_{k+1}$. So one point $x = (a_k)$ makes every C_k true, which contradicts (a). □

Problem 4. *Case $c = 2$.* The single offset is the swap $(x_0, x_1) \to (x_1, x_0)$, a closed run of length 1.

Case $c \geq 3$.

(i) *Defect.* An excursion $x_i \to x_{i'}$ of length ℓ has *defect* $\Delta = \ell - ((i' - i) \bmod c)$. Loops and chords count.

(ii) *Moves.* If token 1 takes it while token 2 stays on C, then token 2 advances ℓ while token 1 lands $\ell - \Delta$ further along C. So the offset changes by $+\Delta$, and by $-\Delta$ if token 2 moves. The interior avoids C, so the only possible collision is at the landing. The move is therefore legal iff the new offset is $\neq 0$.

(iii) *The defects generate $\mathbb{Z}_c$.* A closed walk at x_0 splits into arcs of C and excursions, so its length is $\equiv \sum \Delta \pmod c$. By primitivity these lengths have gcd 1, so they generate $\mathbb{Z}_c$.

(iv) *Connectivity, giving (a).* The offset graph is $\mathrm{Cay}(\mathbb{Z}_c, \pm\{\Delta\})$ with the vertex 0 removed. A connected vertex-transitive graph of degree ≥ 2 is 2-connected (Watkins 1970), so it stays connected.

(v) *Lengths, giving (b).* The chase itself is a closed run of length c. For each Δ, start at an offset $\notin \{0, -\Delta\}$. Token 1 takes the excursion, then token 2 takes it when it arrives. That is a closed run of length $\equiv \Delta$. Since the Δ's generate $\mathbb{Z}_c$, the gcd is 1. □

Problem 5. Token 1 is at v and token 2 at v's predecessor on C. In one step token 1 moves to v's successor on C' (which is $\neq v$, since $|C'| \geq 2$) and token 2 moves onto v. Both are now on C', one apart: a chase on C'. The symmetric step goes back. Chains of cycles sharing vertices connect any two cycles, and a loop in a chain can be skipped, because its neighbours share its vertex. With Problem 4(a), everything is one class. □

Problem 6. The finite algebra B_0 generated by B has atoms that partition Ω. By finite additivity exactly one atom a has $\mu(a) = 1$, and $a \neq \varnothing$. Every element of B_0 contains a or is disjoint from it. So for $\omega \in a$, δ_ω agrees with μ, hence with s_0, on B. □

Problem 7. (a) δ_ω extends s_0 iff $\omega \in E$ for every true E, and $\omega \notin E$ for every false E. A false E has true complement in B, so its constraint is already a true one. Hence the condition is just $\omega \in K(s_0)$. (b) σ-additive two-valued states are Dirac or not. (a) handles the Diracs. The empty-kernel condition is easy to arrange (Navara–Pták does it with even a σ-additive global extension). So the whole difficulty is destroying every non-Dirac σ-additive repair while keeping finite coherence. □

Problem 8. (a) Every window gives each site value 1 with probability $\frac{1}{2}$. (b) Almost surely every window is 01 or 10, so a realising configuration alternates $0, 1, 0, 1, \ldots$ around the ring. That needs L even. (c) The coherence polytope is $\{u \in [0,1]^L : u_i + u_{i+1} \leq 1\}$ and R is the stable-set polytope. The constraint matrix is $I + P$, with $\det(I + P) = 1 - (-1)^L$; its proper square submatrices are path-incidence matrices, with determinants in $\{0, \pm 1\}$. For even L the matrix is totally unimodular, so the polytope is integral and $C = R$. Full proof: archive §"The parity theorem". □

Part IV

Appendix: definitions, notation and background results

Everything the problems use, in one place. The table says what to read for each problem. Entries marked *(standard)* are textbook facts, quoted without proof.

Problem	Read
1	A.1–A.4, A.6
2	C.6
3	A.1–A.4, A.7
4	B.1–B.6, C.1–C.5, D.1–D.3
5	B.1–B.4, B.7, C.1–C.3
6	A.1–A.5, E.1–E.3
7	A.1–A.5, E.1–E.5
8	F.1–F.5

A Measure-theoretic basics

A.1 σ-algebra, Dynkin system. A *σ-algebra* on Ω is a family of subsets containing Ω and closed under complements and countable unions. A *Dynkin system* (or σ-class) only requires closure under complements and *countable disjoint* unions. It need not be closed under intersections. That gap is exactly what makes non-Boolean ("quantum-logic-like") carriers possible.

A.2 Product σ-algebra. On $\Omega = A^I$ (sequences indexed by I, e.g. $I = \mathbb{Z}$ or $\{0, \dots, L-1\}$) it is the smallest σ-algebra making each coordinate $x \mapsto x_k$ measurable. It is generated by the *cylinders* $\{x : x_{k_1} = a_1, \dots, x_{k_m} = a_m\}$. For countable A, every singleton $\{\omega\}$ of $A^{\mathbb{Z}}$ is measurable, being a countable intersection of cylinders.

A.3 Two-valued state; finite vs. σ-additivity. See the shared definitions in Part I. For a two-valued s:

- finite additivity means: of two disjoint sets, at most one is true, and their union is true iff one of them is;
- σ-additivity means the same for countable disjoint families.

Consequences:

- $s(E^c) = 1 - s(E)$;
- s is monotone ($E \subseteq F$, $s(E) = 1 \Rightarrow s(F) = 1$);
- in a σ-algebra, a countable intersection of true sets is true when s is σ-additive.

A.4 Dirac. $\delta_\omega(E) = 1 \iff \omega \in E$. It is always σ-additive. "Two-valued σ-additive = Dirac" holds on any σ-algebra in which a countable partition into singletons (or a countable separating family) is available. Problem 1(a) is an instance.

A.5 Boolean algebra, atoms. A *Boolean algebra* of sets is closed under complement, finite union and finite intersection. A *finite* one is generated by finitely many sets, and its *atoms* (minimal nonempty members) partition Ω. Every member is a union of atoms *(standard)*. "Boolean" = distributive = intersection-closed, in the concrete setting.

A.6 Shift, stationarity. The shift $S : A^{\mathbb{Z}} \to A^{\mathbb{Z}}$ is $(Sx)_k = x_{k+1}$. A state s is *stationary* (*shift-invariant*) if $s(S^{-1}E) = s(E)$ for all E. For Diracs, $\delta_\omega \circ S^{-1} = \delta_{S\omega}$. A *process* is a stationary probability measure on $A^{\mathbb{Z}}$.

A.7 Ultrafilter (for the "why it matters" notes). An *ultrafilter* p on $\mathbb{N}$ is a finitely additive two-valued state on all subsets of $\mathbb{N}$. It is *non-principal* if it is not a Dirac, i.e. every finite set is false. Non-principal ultrafilters exist by Zorn's lemma, and they are *never* σ-additive. The *ordered product* $p \otimes p$ makes $\{(a, b) : a < b\}$ true: "the second coordinate is chosen after, and above, the first". This is the device that fakes an ascending ring.

B Digraphs and languages

B.1 Digraph vocabulary.

- A *walk* is a sequence of vertices joined by arcs. Its *length* is the number of arcs.
- A walk is *closed* if it ends where it starts.
- A *simple cycle* is a closed walk with no repeated vertex other than the endpoints. A *loop* is an arc $v \to v$, a cycle of length 1.

B.2 Strongly connected. Every vertex reaches every other by a walk.

B.3 Period, primitive. For a strongly connected digraph, the *period* is the gcd of all closed-walk lengths, equivalently of all simple-cycle lengths. It is *aperiodic* (*primitive*) if the period is 1. Equivalently, some power of the adjacency matrix is entrywise positive, and then all larger powers are too *(standard; Perron–Frobenius / Wielandt).* In words, for large n there is a walk of length exactly n between any two vertices.

B.4 Language. A relation $\rho \subseteq A \times A$, used as the digraph of legal consecutive symbols. "Primitive language" means its digraph is primitive.

B.5 Excursion, chord, defect. Fix a simple cycle $C = (x_0, \ldots, x_{c-1})$.

- An *excursion* is a walk from some x_i to some $x_{i'}$ whose interior vertices are all off C, other than one of C's own arcs.
- A *chord* is an excursion of length 1, i.e. an arc $x_i \to x_{i'}$ that is not C's arc.
- Its *defect* is $\Delta = \ell - ((i' - i) \bmod c)$: how much longer the excursion is than the arc of C it bypasses, mod c.

B.6 Decomposing a closed walk. A closed walk starting and ending on C splits uniquely into arcs of C and excursions, by cutting at every visit to C.

B.7 Chain-connectivity of cycles *(standard; proved in the repo, session 5).* In a strongly connected digraph, any two simple cycles are joined by a chain of simple cycles in which consecutive cycles share a vertex.

C Two tokens

C.1 Pair graph D. Vertices are ordered pairs (a, b) with $a \neq b$ (the two tokens' positions). There is an arc $(a, b) \to (c, d)$ iff $a \to c$ and $b \to d$ are arcs of ρ and $c \neq d$. A walk in D is a synchronous two-token run that never collides.

C.2 Unordered pair graph $\bar{D}$. The same, with (a, b) and (b, a) identified. A closed walk in $\bar{D}$ returns the tokens to the same *set* of positions, possibly swapped. A *swap* (crossed walk) is a run $(a, b) \rightsquigarrow (b, a)$.

C.3 Chase, offset. On a simple cycle C, a *chase* has both tokens on C, following it, with token 2 a fixed number $j \neq 0$ of positions ahead (the *offset*). It is a closed run of length c. A pair of vertices is *cocyclic* if both lie on one simple cycle, and then it is a chase position.

C.4 SCC, recurrent. A *strongly connected component* (SCC) is a maximal set of mutually reachable vertices. A pair is *recurrent* if it lies on a closed walk of D (equivalently, of $\bar{D}$), i.e. it can return to itself.

C.5 Aperiodic SCC; why it matters. An SCC is aperiodic if the gcd of its closed-walk lengths is 1. The flagship's forcing lemma needs every SCC of $\bar{D}$ that carries a swap to be aperiodic. Otherwise its unsafe ring lengths could sit in a residue class like $3\mathbb{Z}$, which would be a "fork".

C.6 Phase. If walks from a start y to w all have the same length mod c, call it $\lambda(w)$. A token starting at y at time 0 is at w only at times $\equiv \lambda(w)$. A second token can be *out of phase* by e: always at w at times $\equiv \lambda(w) + e$.

D Cyclic groups and Cayley graphs

D.1 $\mathbb{Z}_c$, generation. $\mathbb{Z}_c = \{0, \ldots, c-1\}$ under addition mod c. A set S *generates* $\mathbb{Z}_c$ iff $\gcd(S \cup \{c\}) = 1$. If a set of integers has gcd 1, its residues mod c generate $\mathbb{Z}_c$.

D.2 Cayley graph. $\mathrm{Cay}(\mathbb{Z}_c, \pm S)$ has vertices $\mathbb{Z}_c$ and an edge g—$g \pm s$ for each $s \in S$. It is connected iff S generates $\mathbb{Z}_c$, and it is *vertex-transitive*: every vertex looks the same.

D.3 Watkins' theorem *(standard; Watkins 1970).* A connected vertex-transitive graph of degree d has vertex connectivity at least $\frac{2}{3}(d+1)$. In particular, for $d \geq 2$ removing one vertex leaves it connected.

E Quantum-logic side (σ-essential lane)

E.1 Concrete σ-orthoposet. A family $\mathcal{L}$ of subsets of Ω, containing Ω, closed under complement ($E^{\perp} = E^c$) and under countable unions of pairwise disjoint members. This is a Dynkin system (A.1) viewed as a "logic". Disjoint sets play the role of *orthogonal* events.

E.2 $\perp$-closed sub-orthoposet; local pattern. A finite $B \subseteq \mathcal{L}$ closed under complement: the finitely many events actually being reported. A *two-valued state on B* (local pattern) s_0 assigns 0/1 with $s_0(E^c) = 1 - s_0(E)$ (and additivity on disjoint pairs within B).

E.3 Finitely coherent. s_0 extends to some *finitely additive* two-valued state on all of $\mathcal{L}$.

E.4 Kernel. $K(s_0) = \bigcap\{E \in B : s_0(E) = 1\}$, the points compatible with every reported "yes".

E.5 σ-essential contextual state. A local pattern that is finitely coherent but extends to *no* σ-additive two-valued state on $\mathcal{L}$: coherence that only finite additivity can honour. The Navara–Pták example (1983) has empty kernel yet a σ-additive extension, which shows that "empty kernel" alone is not enough.

F Commensurability side (flagship)

F.1 Ring protocol, windows. Sites $\mathbb{Z}_L$ around a ring. Each adjacent pair $\{i, i+1\}$ is a *window*, reported as a probability table over the legal pairs of ρ.

F.2 Coherent (EA) vs. realisable (PR). The window data are *coherent* (empirically adequate) if overlapping windows give the same single-site marginals. They are *realisable* if some probability measure on legal ring configurations has exactly these window marginals. *Contextual* means coherent but not realisable. A protocol is *commensurable* if every coherent datum is realisable.

F.3 Polytopes C, R. Coherent data form a polytope C, and realisable data the convex hull $R \subseteq C$ of the deterministic configurations. Commensurable $\iff C = R$.

F.4 Stable sets. In a graph, a *stable (independent) set* has no two adjacent vertices.

- STAB is the convex hull of their indicator vectors.
- FSTAB $= \{u \in [0,1]^V : u_i + u_j \leq 1 \text{ on edges}\}$ is its fractional relaxation.
- For the golden mean on a ring, $C \cong \mathrm{FSTAB}(C_L)$ and $R \cong \mathrm{STAB}(C_L)$, with $u_i = \Pr[x_i = 1]$.

F.5 Total unimodularity *(standard; Hoffman–Kruskal).* A matrix is *totally unimodular* if every square submatrix has determinant in $\{0, \pm 1\}$. Then $\{u : Mu \leq b,\, u \geq 0\}$ has integral vertices for every integral b. Also used:

- $\det(I + P) = \prod_k (1 + \omega^k) = 1 - (-1)^L$ for the cyclic shift P, whose eigenvalues are the L-th roots of unity ω^k;
- path-incidence matrices have determinants in $\{0, \pm 1\}$ (triangularise along the path).